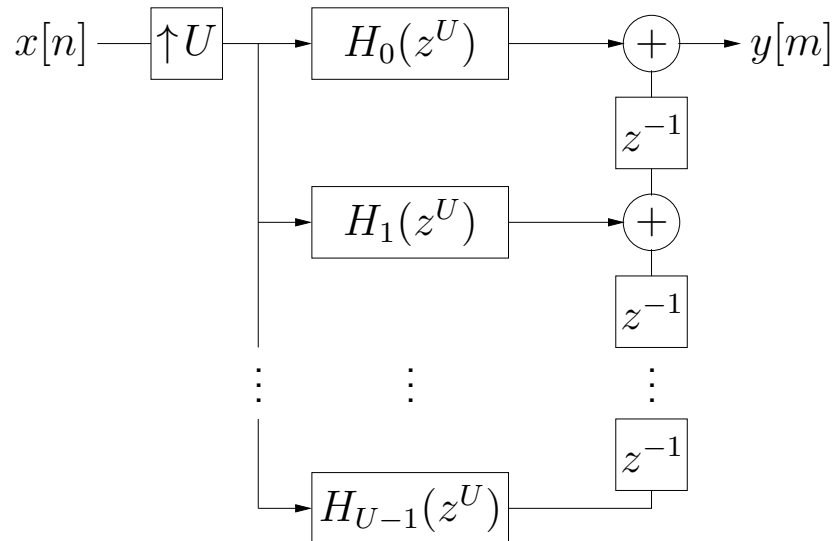
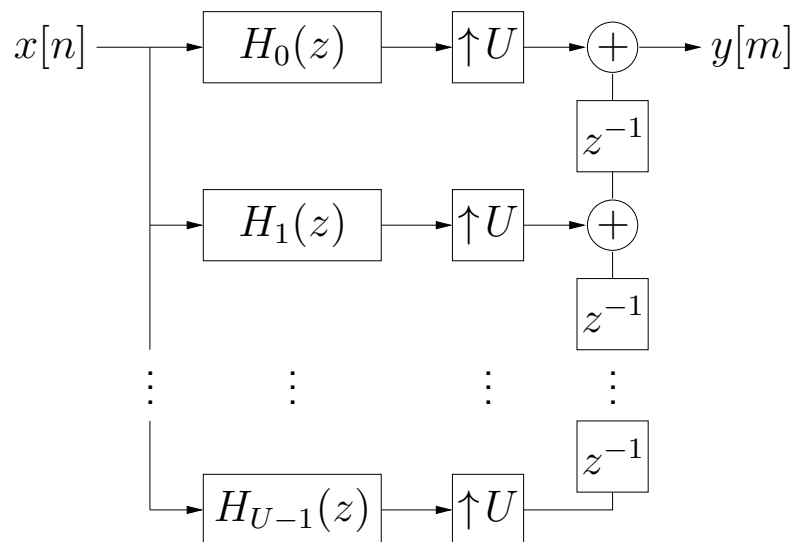


Polyphase Interpolation (cont.):

- With $H(z) = \sum_{p=0}^{U-1} H_p(z^U) z^{-p}$, interpolation becomes



- Moving $\uparrow U$ into the branches and applying the Noble identity for upsampling, we get

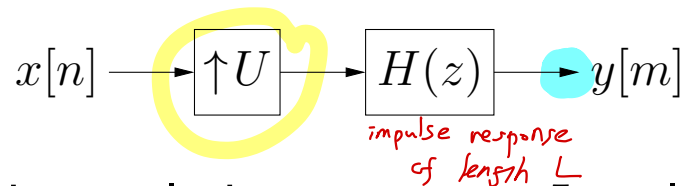


where now the filters don't operate on upsampled signals!

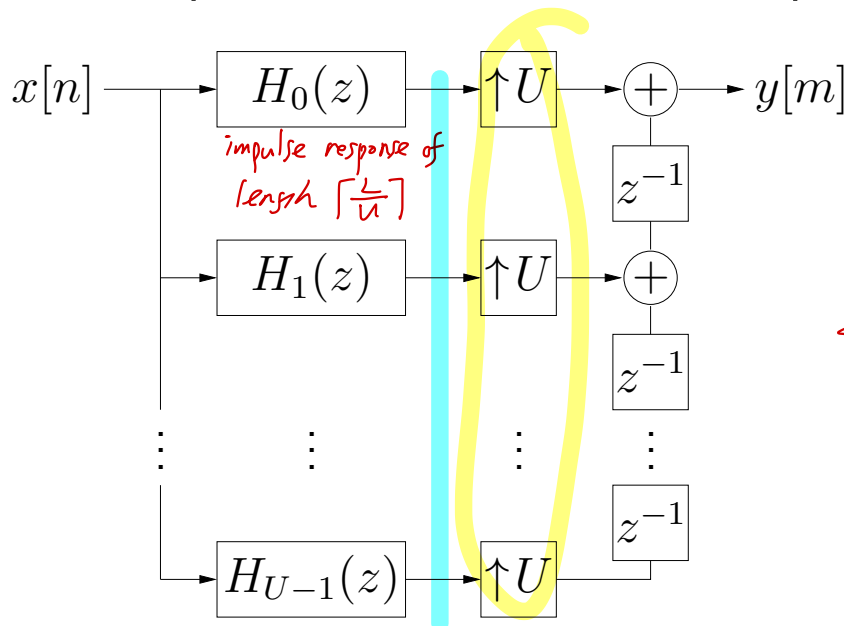
- The “ladder” of upsamplers and delays on the right performs parallel-to-serial conversion.

Computational Savings of Polyphase Interp:

- Suppose that $H(z)$ is a length- L FIR filter.
- Standard interpolation consumes L multiplies per output:



- Polyphase interpolation consumes L multiplies per input:

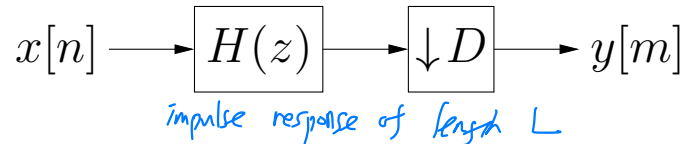


$$\begin{aligned}
 &44.1 \text{ kHz} \\
 &\rightarrow 48 \text{ kHz} \\
 &U = 160 \\
 &L = 144
 \end{aligned}$$

- Since there are U times as many outputs as inputs, polyphase interpolation uses a factor-of- U fewer multiplies.

Polyphase Decimation:

- The standard decimation procedure

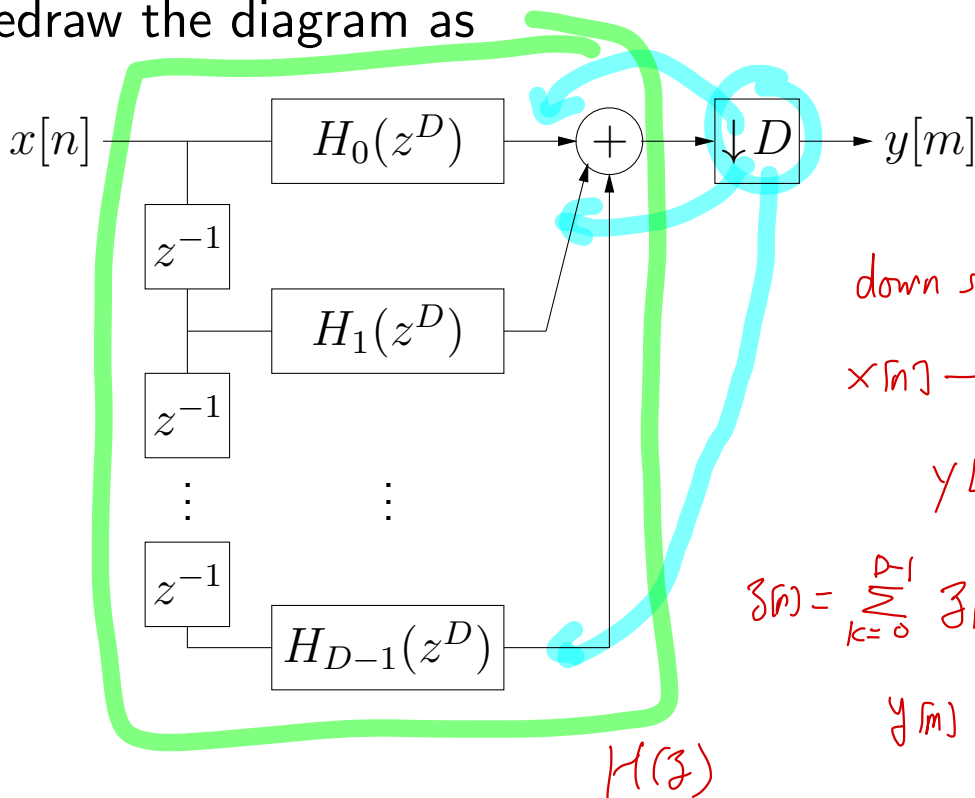


is computationally inefficient because it discards most of the filter outputs.

- Using the Noble identities, we can design an efficient decimator.
- As before, we can use the polyphase decomposition

$$h_p[m] \triangleq h[mD + p] \Rightarrow H(z) = \sum_{p=0}^{D-1} H_p(z^D) z^{-p}$$

to redraw the diagram as



down sampling

$$x[n] \rightarrow \boxed{\downarrow D} \rightarrow y[m]$$

$$y[m] = x[mD]$$

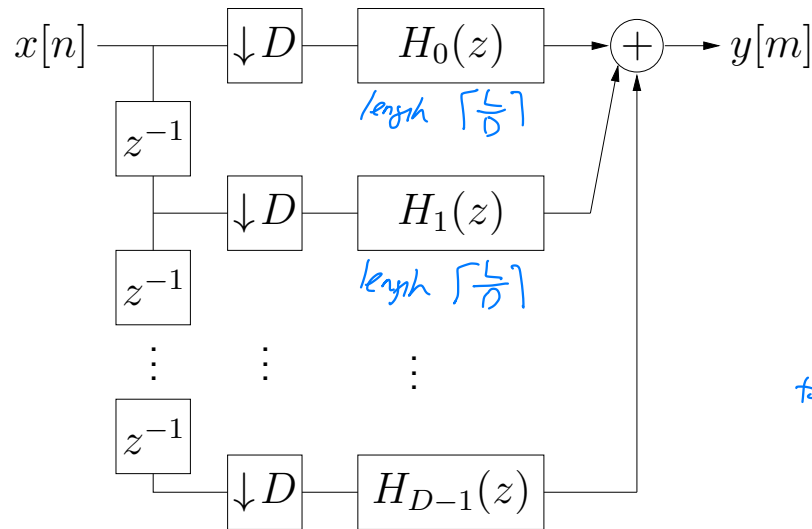
$$z[n] = \sum_{k=0}^{D-1} z_k[n] \rightarrow \boxed{\downarrow D} \rightarrow y[m]$$

$$y[m] = z[mD]$$

$$= \sum_{k=0}^{D-1} z_k[mD]$$

Polyphase Decimation (cont.):

- Moving $\downarrow D$ into the branches and applying the Noble identity for downsampling, we get



total # multi;
 $= D \cdot \lfloor \frac{L}{D} \rfloor \approx L$

where no filter outputs are discarded!

- The “ladder” of delays and downsamplers on the left performs serial-to-parallel conversion.

Computational Savings:

- Polyphase decimation consumes a factor-of- D fewer multiplies than standard decimation since...
 - Standard consumes L multiplies per input
 - Polyphase consumes L multiplies per output
 - There are D times as many inputs as outputs.

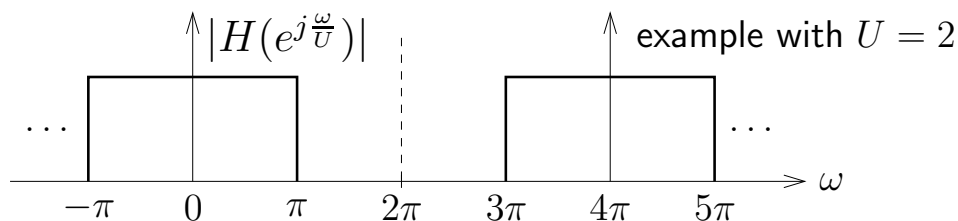
Polyphase Filters as Allpass Delays:

- Consider the *ideal* linear-phase interpolation lowpass filter:

$$H(e^{j\omega}) = \begin{cases} Ue^{-j\omega\ell} & \omega \in (-\frac{\pi}{U}, \frac{\pi}{U}] \\ 0 & \omega \in (-\pi, -\frac{\pi}{U}] \cup (\frac{\pi}{U}, \pi] \end{cases}$$

- For each p , the polyphase filter $h_p[m] = h[mU + p]$ is a U -downsampled version of p -advanced $h[n]$. Thus

$$\begin{aligned} H_p(e^{j\omega}) &= \frac{1}{U} \sum_{k=0}^{U-1} V(e^{j\frac{\omega-2\pi k}{U}}) \quad \text{where } V(e^{j\omega}) = e^{j\omega p} H(e^{j\omega}) \\ &= \frac{1}{U} \sum_{k=0}^{U-1} e^{j(\frac{\omega-2\pi k}{U})p} H(e^{j\frac{\omega-2\pi k}{U}}) \\ &= \frac{1}{U} e^{j\frac{\omega}{U}p} H(e^{j\frac{\omega}{U}}) \quad \text{for } |\omega| \leq \pi \\ &= e^{-j\frac{\ell-p}{U}\omega} \quad \text{for } |\omega| \leq \pi. \end{aligned}$$

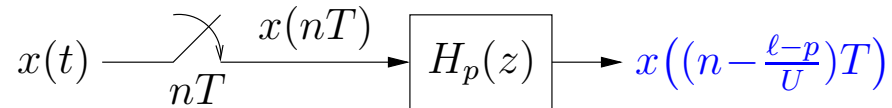


- Since $|H_p(e^{j\omega})| = 1 \quad \forall \omega$, the polyphase filters are *allpass*.
- Since $\phi(\omega) = -\frac{\ell-p}{U}\omega$ for $H_p(e^{j\omega})$, its group delay is

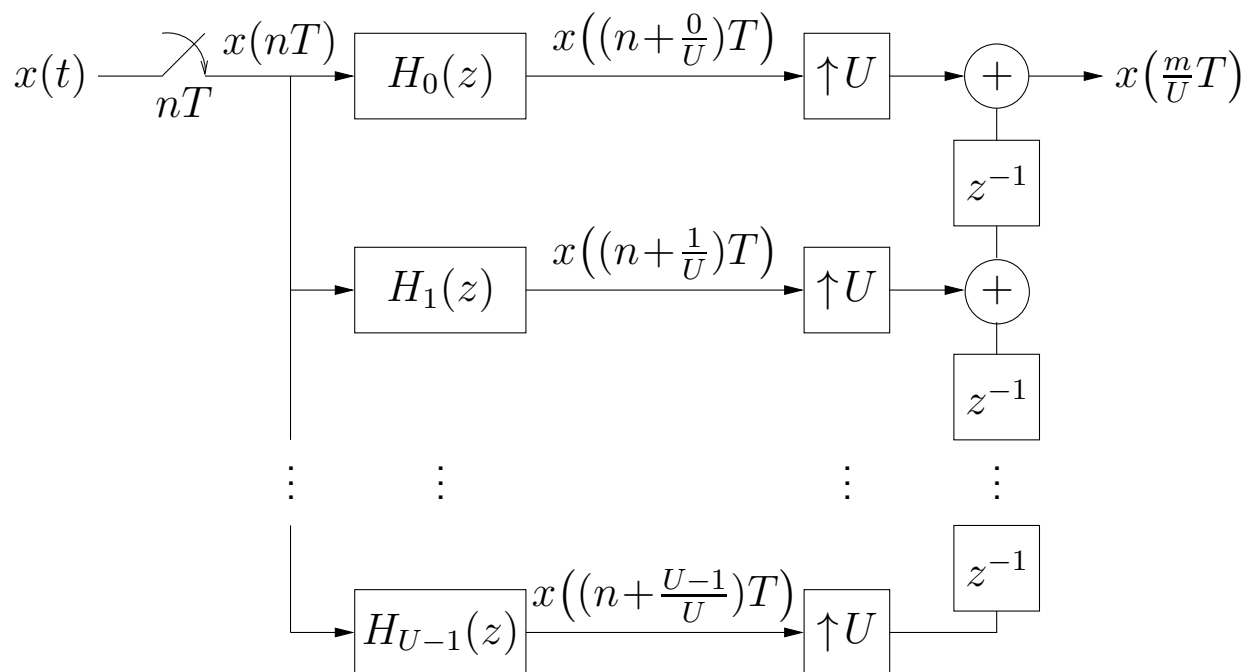
$$g(\omega) = -\frac{d\phi(\omega)}{d\omega} = \frac{\ell-p}{U},$$

Polyphase Filters as Allpass Delays (cont.):

- So the ideal polyphase filter delays by $\frac{\ell-p}{U}$ samples!



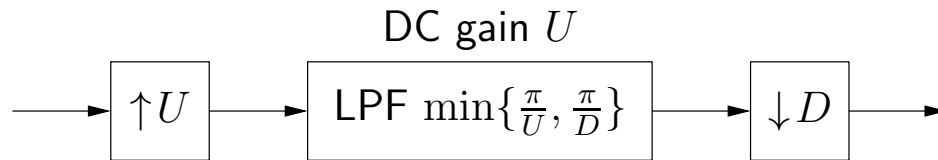
- Thus, the polyphase interpolator with ideal polyphase filters implements (using $\ell = 0$ for illustration)



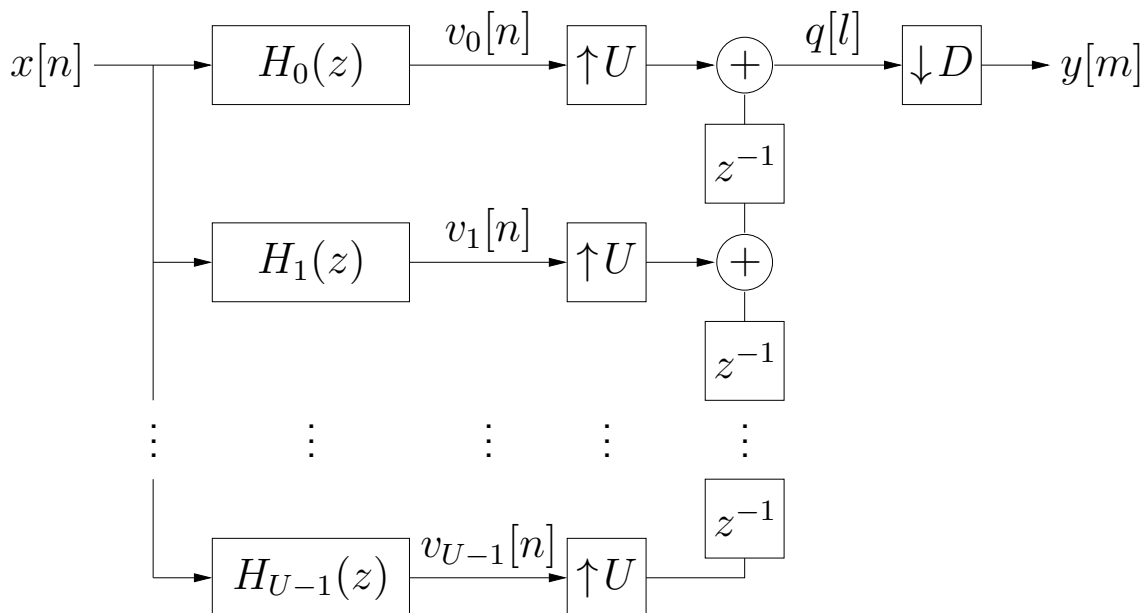
which can be recognized as ideal interpolation.

Polyphase Resampling:

- Recall that we can resample by rational rate U/D using



- Suppose that we implemented the upsampler/LPF pair using polyphase:



This is still inefficient: $\downarrow D$ throws away computations.

- Idea:
 - The ladder of upsamplers/delays performs parallel-to-serial conversion.
 - Compute only the branch that is needed at time m .

Polyphase Resampling (cont.):

- Given the “parallel-to-serial” interpretation, the input to the downsampler, $q[l]$, takes the form

$$\dots, v_0[0], v_1[0], v_2[0], \dots, v_{U-1}[0],$$

$$v_0[1], v_1[1], v_2[1], \dots, v_{U-1}[1],$$

$$v_0[2], v_1[2], v_2[2], \dots, v_{U-1}[2], \dots$$

leading to the observation that

$$q[l] = v_{\langle l \rangle_U} \left[\left\lfloor \frac{l}{U} \right\rfloor \right]$$

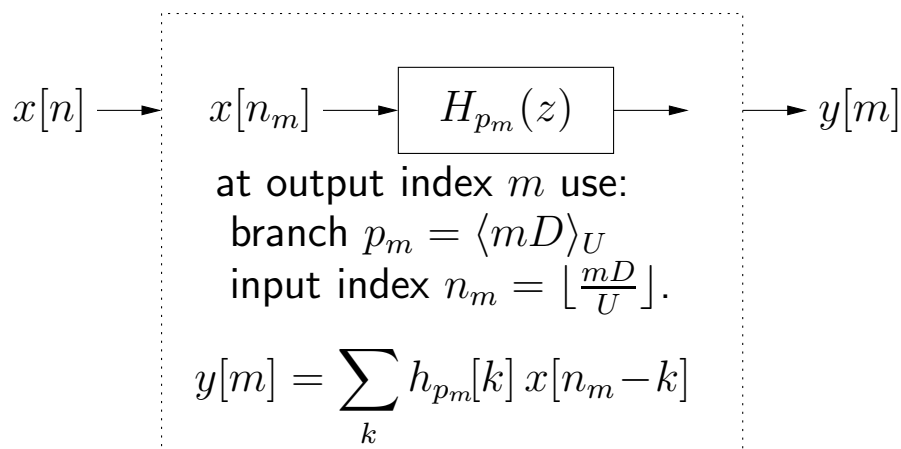
channel index sample index

$$y[m] = q[mD]$$

$$= v_{\langle mD \rangle_U} \left[\left\lfloor \frac{mD}{U} \right\rfloor \right]$$

$$= \sum_k h_{\langle mD \rangle_U}[k] x \left[\left\lfloor \frac{mD}{U} \right\rfloor - k \right]$$

- We can summarize this procedure as follows:



- Computational Savings: standard resampling consumes $UD\times$ as many mults as polyphase.